Practice workbook · Explain how inhibitors or a mutation change enzyme activity

Competitive and non-competitive inhibitors, drugs such as aspirin acting as inhibitors, and mutations that change an enzyme's shape.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ENZYME-INHIBITOR-MUTATION

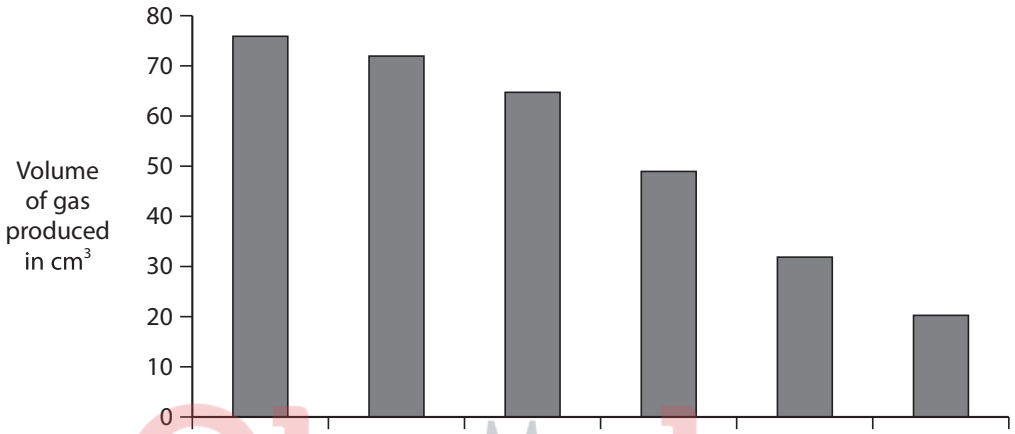
1. Question 3 · part b · 2 marks

June 2023 Paper 1

Worked-example retry · Attempt part b only.

3 A student investigates a reaction. The reaction is catalysed by an enzyme and produces a gas. The student collects the gas for 10 minutes and measures the volume of the gas. The student repeats this every 10 minutes for one hour.

The bar chart shows the student's results.



(a) (i) Add a suitable labelled scale to the *x*-axis.

(2)

(ii) Estimate the total volume of gas produced after 30 minutes.

(2)

volume of gas = ... cm^3

(iii) Describe the trend shown by the results.

(3)

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1. Question 3 · part b · 2 marks

June 2023 Paper 1

Worked-example retry · Attempt part b only.

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(iv) Explain the student's results obtained in the investigation.

(3)

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(b) Explain the effect on the results if the student had added a competitive inhibitor at the start of the investigation.

(2)

(Total for Question 3 = 12 marks)

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1. Question 3 · part b · 2 marks · Working and answer

Working and answer

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2. Question 5 · part b.iii · 2 marks

November 2023 Paper 1

Further practice · Attempt part b.iii only.

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5 DNA codes for the production of proteins in cells.

(a) The DNA code is read, three bases at a time. There are four different bases.

Calculate how many different three base codes are available.

(2)

number of codes = ...

(b) (i) Explain how a mutation in DNA can cause a change in the phenotype of a cell.

(3)

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(ii) State **one** factor that can cause mutation.

(1)

(iii) A mutation can change the shape of an enzyme molecule.

Explain why this may change how the enzyme works.

(2)

2. Question 5 · part b.iii · 2 marks

November 2023 Paper 1

Further practice · Attempt part b.iii only.

(iv) State why some mutations have no effect on the phenotype of an organism.

(1)

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(Total for Question 5 = 9 marks)

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Working and answer

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3. Question 9 · part c · 4 marks

June 2019 Paper 2

Further practice · Attempt part c only.

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9 Aspirin is a painkiller, but can also be prescribed to people who are at risk of having a stroke or a heart attack.

Aspirin dissolves blood clots and is also an enzyme inhibitor, which reduces the risk of more blood clots forming in blood vessels.

(a) (i) Describe how a blood clot forms in a blood vessel.

(3)

(ii) Explain how a blood clot in the coronary artery increases the risk of a heart attack.

(3)

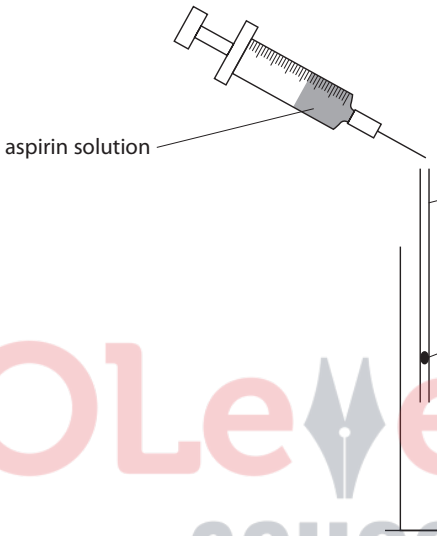
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3. Question 9 · part c · 4 marks

June 2019 Paper 2

Further practice · Attempt part c only.

(b) A student uses this apparatus to investigate the effect of aspirin on the rate of blood flow through blood vessels.



The student uses this method.

- dissolve one aspirin tablet in water
- use a syringe to pass the aspirin solution through a straw containing a blood clot
- record the total volume of aspirin solution in the measuring cylinder every five minutes

The student repeats the method with solutions formed from two aspirin tablets and then from three aspirin tablets.

(i) Give a control variable for this investigation.

(1)

(ii) Describe a suitable control test for this investigation.

(1)

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3. Question 9 · part c · 4 marks

June 2019 Paper 2

Further practice · Attempt part c only.

The table shows the student's results.

Time in minutes	Total volume of aspirin solution in the measuring cylinder in cm^3		
	1 aspirin tablet	2 aspirin tablets	3 aspirin tablets
5	0	0	0
10	0	0	0
15	0	0	1
20	0	1	2
25	1	1	3
30	1	2	4
35	2	4	6
40	5	7	9
45	10	12	16
50	17	19	22
55	25	27	31
60	35	37	39

(iii) Explain the pattern shown by the results.

(3)

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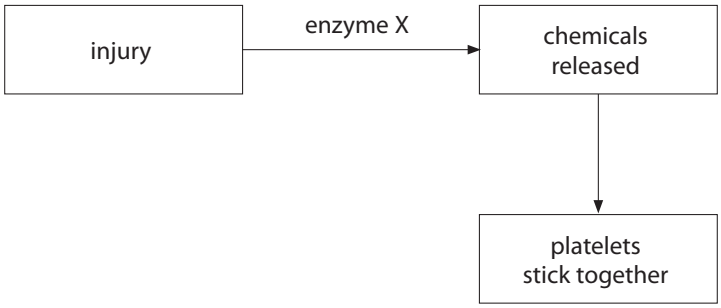
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3. Question 9 · part c · 4 marks

June 2019 Paper 2

Further practice · Attempt part c only.

(c) The diagram shows a pathway involved in blood clotting.



Aspirin is an inhibitor of enzyme X.

Explain how the inhibition of enzyme X reduces the formation of blood clots.

(4)

(Total for Question 9 = 15 marks)

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3. Question 9 · part c · 4 marks · Working and answer

Working and answer

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4. Question 2 · part b.iii · 2 marks

June 2022 Paper 2R

Further practice · Attempt part b.iii only.

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2 A student investigates the effect of a competitive inhibitor on the action of the enzyme phosphatase.

This is the reaction catalysed by the enzyme.

phenolphthalein phosphate ⟶ phenolphthalein + phosphate

Phenolphthalein turns pink when an alkali is added.

This colour can be measured using a colorimeter. The more intense the colour, the higher the reading on the colorimeter.

This is the student's method.

- prepare six different concentrations of phenolphthalein phosphate solution
- place 5.0 cm^3 of each solution into a different test tube
- add 1.0 cm^3 of a competitive inhibitor and 1.0 cm^3 of the enzyme phosphatase into each test tube
- wait 30 minutes, then add 1.0 cm^3 of an alkali to each tube
- measure the intensity of the pink colour using a colorimeter

(a) (i) Give a suitable control for this investigation. **(2)**

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(ii) State three variables that should be kept constant in this investigation. **(3)**

1 .....

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2 .....

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3 .....

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4. Question 2 · part b.iii · 2 marks

June 2022 Paper 2R

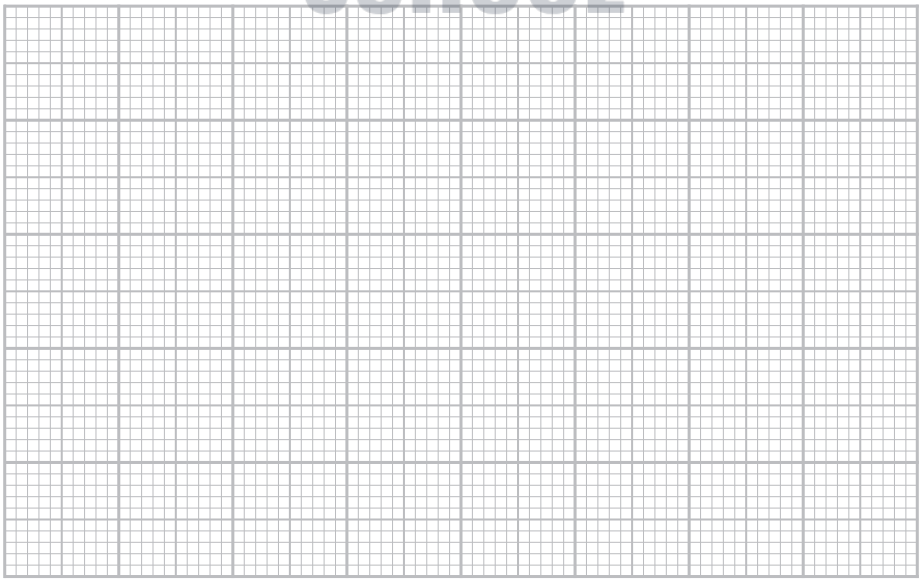
Further practice · Attempt part b.iii only.

(b) The table shows the student's results.

Concentration of phenolphthalein phosphate solution in arbitrary units	Intensity of pink colour in arbitrary units
0.05	0.20
0.10	0.30
0.20	0.47
0.40	0.65
0.60	0.79
0.80	0.91

(i) Plot the results of the investigation. **(3)**

(ii) Draw a curve of best fit. **(1)**



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4. Question 2 · part b.iii · 2 marks

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Further practice · Attempt part b.iii only.

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(iii) Add a second curve to show the results that would be obtained if the competitive inhibitor had not been added.

(2)

(c) Describe how the results of the investigation would be different if a non-competitive inhibitor had been used instead of a competitive inhibitor.

(2)

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(Total for Question 2 = 13 marks)

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Working and answer

5. Question 2 · part c · 2 marks

June 2022 Paper 2R

Further practice · Attempt part c only.

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This is the reaction catalysed by the enzyme.

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Phenolphthalein turns pink when an alkali is added.

This colour can be measured using a colorimeter. The more intense the colour, the higher the reading on the colorimeter.

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- wait 30 minutes, then add 1.0 cm^3 of an alkali to each tube
- measure the intensity of the pink colour using a colorimeter

(a) (i) Give a suitable control for this investigation. **(2)**

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(ii) State three variables that should be kept constant in this investigation. **(3)**

1 .....

2 .....

3 .....

5. Question 2 · part c · 2 marks

June 2022 Paper 2R

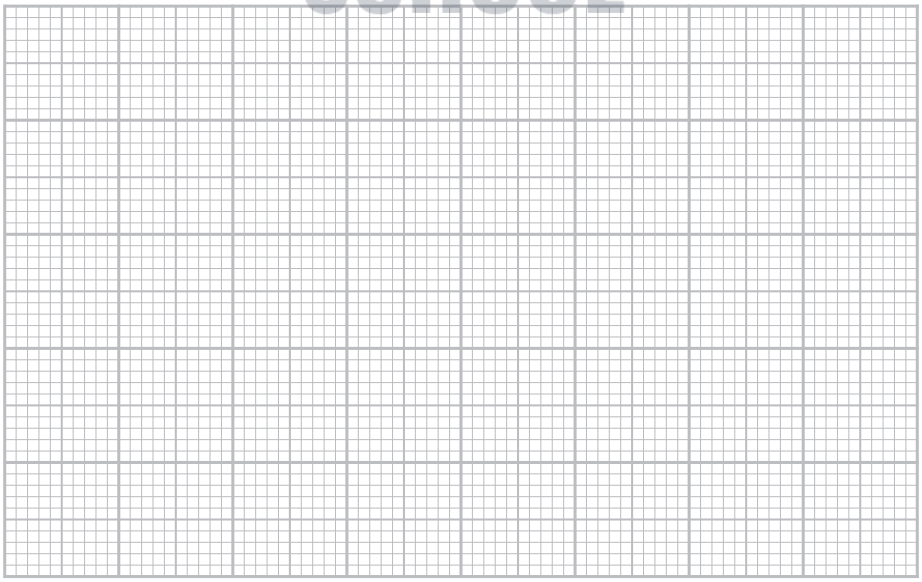
Further practice · Attempt part c only.

(b) The table shows the student's results.

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0.05	0.20
0.10	0.30
0.20	0.47
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0.60	0.79
0.80	0.91

(i) Plot the results of the investigation. **(3)**

(ii) Draw a curve of best fit. **(1)**



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5. Question 2 · part c · 2 marks

June 2022 Paper 2R

Further practice · Attempt part c only.

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(iii) Add a second curve to show the results that would be obtained if the competitive inhibitor had not been added.

(2)

(c) Describe how the results of the investigation would be different if a non-competitive inhibitor had been used instead of a competitive inhibitor.

(2)

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(Total for Question 2 = 13 marks)

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Working and answer